

SkillClaw

Collective Skill Evolution for Multi-User LLM Agent
Ecosystems

arxiv.org/abs/2604.08377



Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Complementary Views from Diverse Contexts

Different users exercising the same skill under varied conditions reveal both working conditions and failure modes.

- OpenClaw
- CoPaw
- IronClaw
- PicoClaw
- ZeroClaw
- NanoClaw
- NemoClaw

A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.

Trajectory Signal Diversity



■ Success signals ■ Failure signals



Three Properties That Distinguish SkillClaw

Collective Evolution

01

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem.

Fully Automatic

02

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention.

Agentic Paradigm

03

Skill updates are produced through open-ended reasoning rather than predefined update rules.

Implications for Continuously Improving Agent Systems

- System-level knowledge accumulation replaces individual rediscovery
- Cross-user diversity strengthens behavioral boundary detection
- Conservative verification prevents regressions while enabling improvement
- Compatible with WildClawBench evaluation framework